Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Real-Time CTM-Centric Precision Oncology Platform for Early Micro-Metastasis Prediction and Targeted Therapeutic Intervention			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	R&D / Engineering Services	Design Information Integration Management / Collaboration System Performance Enhancement Technologies
Concept of Proposed Technology	The proposed technology is an **integrated CTM (circulating tumor microembolus)–centric precision oncology platform** to enable **early prediction, functional characterization, and therapeutic intervention of micro-metastatic disease** in patients with non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC). Metastasis is the primary cause of cancer-related mortality. Studies demonstrate that CTMs, multicellular clusters derived from circulating tumor cells (CTCs), exhibit significantly higher metastatic potential than single CTCs, mediated by cooperative survival signaling, immune evasion, and enhanced colonization capacity(Aceto et al., 2014; Szczerba et al., 2019). However, current liquid biopsy approaches largely rely on static enumeration or endpoint molecular profiling and cannot capture the **dynamic biological processes driving micro-metastatic progression**(Castro-Giner & Aceto, 2020). The proposed platform integrates the following core components: • **CTC-derived organoid chip technology**, creating a three-dimensional biomimetic microenvironment supporting CTC clustering and CTM formation • **PICASI-based real-time functional analysis**, enabling non-invasive, longitudinal monitoring of CTM initiation, aggregation kinetics, structural stability, and morphological evolution • **Time-resolved multi-omics profiling**, including next-generation sequencing (NGS) based DNA- and RNA-seq and phosphoproteomic antibody chip analysis of isolated CTMs • **AI-based multi-omics integration**, modeling CTM biological states and predicting metastatic risk and therapeutic responsiveness • ***Ex vivo* functional validation**, using CTM organoids to evaluate drug repurposing candidates and genome-targeted therapeutics • By using the temporal dynamics of **CTC/CTM formation** as a quantitative functional			

	biomarker, the platform complements tumor-centric mutation analysis by enabling **earlier detection and potential interception of metastatic processes** that underlie cancer mortality.
Rationale for Joint R&D Project	Existing clinical strategies for precision oncology remain centered on primary tumor genomics and established metastases. These approaches are fundamentally limited in their ability to predict or intercept micro-metastatic disease at an early, therapeutically actionable stage (Yuan et al., 2022; Lin et al.,2021). CTMs represent a biologically validated precursor of metastatic dissemination, yet current technologies lack the ability to: • Reproduce CTM formation in a physiologically relevant *ex vivo* system • Monitor CTM formation and evolution in real time • Functionally characterize CTM-specific molecular and signaling dependencies • Translate CTM biology into actionable therapeutic selection(Castro-Giner & Aceto, 2020; Rozenberg et al., 2023) Addressing these challenges requires **integrating organoid chip and other advanced platform technologies, translational cancer biology, multi-omics analytics, and clinical validation;** achieving this scope and scale necessitates a multi-institutional, multidisciplinary consortium(Liu et al., 2021). International joint R&D is essential to: • Access clinically annotated NSCLC and SCLC patient cohorts available through the Korean SME Partner • Independently validate CTM-derived biomarkers (somatic mutations) as therapeutic targets • Ensure global credibility, reproducibility, and translational relevance • Align platform development with international regulatory and clinical research standards The joint framework supports iterative co-development, by integrating CTM modeling, functional analysis, prioritization of lead candidate (target genes and mutation, and drug compounds) and platform optimization. This integration allows insights from each component to guide refinements in the others and produces cross-cutting synergies not achievable through outsourcing or single-institution efforts.
Key Words	1) Circulating Tumor Microemboli (CTM), 2) Real-Time Functional Analysis (PICASI), 3) Lung Cancer Circulating Tumor Cell (CTC)-Derived Organoid Platform, 4) Multi-Omics Precision Oncology, 5) Metastasis-Targeted Therapeutic Development

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3-4)	**End of the Project**	Level (5-6)
R&D Objectives	**Overall Goal** To develop and clinically validate a **CTM-centric precision oncology platform** enabling early micro-metastasis prediction, functional therapeutic selection, and targeted anti-metastatic intervention, advancing the technology from **TRL 3–4 to TRL 5–6**. **Objectives** • Establish a PICASI-enabled CTC organoid platform for real-time CTM formation analysis • Define CTM formation dynamics as quantitative biomarkers of micro-metastatic risk • Characterize CTM-specific genomic alterations and signaling dependencies using multi-omics profiling • Develop AI-based multi-omics integration models for drug repurposing and risk stratification • Validate patient-specific drug responsiveness through *ex vivo* CTM organoid assays			

	• Advance genome-targeted anti-metastatic strategies using shRNA- and mRNA-based therapeutics
R&D Target Specifications	1) Real-Time CTM Analysis • Temporal resolution: ≤ 5-minute intervals for ≥ 72 hours • Detection sensitivity: CTM detection from ≥ 5 CTCs per assay • Reproducibility: CV ≤ 15% • Outputs: Initiation time, aggregation rate, stability index 2) CTC Organoid Chip Performance • Long-term viability: ≥ 70% over 7–14 days • CTM formation efficiency: ≥ 60% of CTC-positive samples • Throughput: ≥ 8–16 patient samples per run 3) Multi-Omics Profiling • DNAseq (whole exome) depth: ≥ 200x to 400× (CTM DNA) • CNV resolution: ≥ 1 Mb • Bulk mRNA-seq raw reads: 30 to 60 million paired reads (15 to 30M read1 and 15 to 30M read2) • Phospho-proteomic coverage: ≥ 30 to 50 phospho-targets 4) Machine learning-Based Integration • Drug response prediction: AUC ≥ 0.80 • Micro-metastatic risk stratification accuracy: ≥ 75% • Explainability: Top 10–20 CTM-driving features identified 5) Drug Repurposing and Validation • Drug library size: ≥ 20 compounds • Hit rate: ≥ 10–20% with ≥ 50% CTM suppression • Turnaround time: ≤ 21 days from blood collection 6) Genome-Targeted Therapeutics • Target genes identified: ≥ two • Gene knockdown efficiency: ≥ 70% • Functional impact: ≥ 50% reduction in CTM formation or stability 7) Clinical Validation • Patient samples: 30 to 50 total cases (combined NSCLC and SCLC) • Clinical correlation: $p < 0.05$ with progression-free survival or treatment response (For several quantitative aims, ranges are provided to accommodate expected variation in NSCLC and SCLC availability and in CTC/CTM recovery and yield)
R&D Plan	The project follows a **phased, milestone-driven 36-month plan**. Phase 1: Platform Establishment and Real-Time CTM Monitoring • Optimize CTC organoid chip conditions • Integrate PICASI real-time CTM analysis • Define baseline CTM formation metrics Phase 2: CTM Isolation and Multi-Omics Characterization • Time-resolved CTM isolation, nucleic acids (DNA, RNA) and protein purification • NGS-based whole exome DNA-seq (WES) and bulk mRNA-seq (whole transcriptome) and phosphoproteomic profiling • Identification of metastasis-driving pathways Phase 3: AI-Based Integration and Drug Repurposing • Multi-omics data harmonization (WES, bulk mRNA-seq, and phosphoproteomics)

	• Machine learning (ML)-based risk and drug response modeling • Candidate (target genes and mutations) therapeutic prioritization Phase 4: *Ex Vivo* Drug Sensitivity Validation • Functional CTM suppression and viability testing • ML-based prediction of risk stratification and drug repurposing • Patient-specific therapeutic stratification Phase 5: Genome-Targeted Therapeutic Development • Selection of metastasis-driving genes • Design and validation of the metastasis-driving target gene(s) using RNAi (shRNA or siRNA) Phase 6: Translational and International Validation • Clinical correlation of the identified biomarkers with patient outcomes (overall survival (OS), progression-free survival (PFS), and time to tumor recurrence) and demographic and clinical covariates (e.g., age at diagnosis, gender, disease state, smoking history/status, histology (NSCLC vs. SCLC))Analytical and clinical validation of biomarkers (target genes /mutations) and workflows • Preparation of regulatory dossiers and multi-center scale-up

4. R&R (Roles and Responsibilities):

Expected R&R of Research Institute	**Aims of the Johns Hopkins University (JHU) Biomedical Engineering (BME) research team** **Aim 1.** Molecular characterization of small cell lung cancer (SCLC) CTCs and CTMs: Bioinformatic analysis of DNAseq (whole exome seq (WES)) and bulk RNAseq (transcriptomics) data • Call somatic single-base substitutions (SBS) from WES; cross-validate/augment calls using bulk mRNA-seq, with explicit handling of RNA-specific biases. • Assess how changes in mutation presence (Variant allele frequency (VAF) over cancer treatments correlate with transcriptional changes. • Quantify mutation dynamics: estimate variant allele frequency (VAF) per SBS per timepoint; classify variants as emerging, persistent, or disappearing. • Associate temporal changes in mutation VAF with within-sample gene-level expression changes and pathway analysis (differential expression, gene-set enrichment). **Aim 2**. Enhancement of algorithms for clinical application through validation of longitudinal analysis of mutation and gene expression profiles of SCLC CTCs and CTMs: • Model time-dependent relationships using mixed-effects or joint longitudinal-survival models to relate molecular dynamics (mutation and gene expression changes) to clinical outcomes (e.g. survival days after diagnosis) • Validate and refine predictive algorithms using time series molecular profiling data specifically RNAseq from circulating tumor cells (CTCs) and circulating tumor microemboli/clusters (CTMs) in small cell lung cancer (SCLC). • Collaboration is available with Joel Bader, PhD (JHU BME Professor) to support this algorithm enhancement, including development of data-mining pipelines (Bayesian mixture models, Artificial Neural Networks) tailored for longitudinal mutation datasets from SCLC CTCs and CTMs. **Aim 3.** Development of an extracellular matrix (ECM)-mimetic 3D cancer invasion model of CTC-derived organoids and molecular characterization of these organoids. - ***Note***: *Implementation of this Aim 3 is* ***conditional*** *upon:* (a) institutional approvals between Johns Hopkins University and the Korean company, (b) execution of a material transfer agreement, and (c) demonstration of successful cryopreserved (liquid nitrogen) shipment that preserves acceptable viability of circulating tumor cells (CTCs) and circulating tumor microemboli (CTMs) from Korea to USA. If any of these conditions are unmet or it takes unreasonably long time that does not fit within the timeline of this grant, the following experiments will not be conducted. Instead, the JHU team will prioritize the bioinformatics analyses of the sequencing data and development of AI algorithms described above (***Aims 1 and 2***).

	• Build an ECM-mimetic 3D CTC/CTM tumoroid invasion model by forming size-controlled CTC/CTM aggregates in AggreWell plates and embedding them in fibrillar collagen I (isotropic or aligned) to capture collective invasion in a tunable matrix. • Enable scalable, clinically relevant screening by adapting the assay to multi-well workflows for NSCLC/SCLC comparisons and medium-throughput anti-invasive drug testing with automated imaging/analysis readouts. • Molecular characterization of CTC organoids on the 3D invasion platform using DNAseq and bulk RNAseq.
Expected R&R of Korean SME Partner	The Korean SME partner will lead **platform engineering and system integration**, including: • Development of the CTC organoid–PICASI platform • Assay standardization and SOP development • AI-based multi-omics analytics • Drug repurposing and functional validation workflows • Data management and regulatory readiness • Commercialization strategy development See ***R&D plan (Phases 1 to 6)*** in Section ***3. Research and Development Objectives & Plan*** for details.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Scientific and Technological Impact • Establishment of a **first-in-class CTM-centric precision oncology platform** • Identification of functional biomarkers (target genes, somatic mutations) for early micro-metastasis detection and prediction Early Detection and Functional Modeling of Micrometastatic Disease: • Translating longitudinal molecular signatures (somatic mutations and gene expression) in CTCs/CTMs across treatment into clinically actionable predictions of treatment response, survival, and tumor recurrence to enable earlier detection of micrometastatic disease in SCLC. • Applying CTC-derived organoids as an *in-vivo*-like functional platform for early detection and modeling of micrometastatic disease. Clinical Impact • Enabling earlier intervention in metastatic progression • Improved therapy selection and prediction of treatment response • Providing evidence for clinical utility by correlating biomarkers with clinical outcomes (overall survival, treatment response) and clinical covariables (e.g., age at diagnosis, gender, smoking history/status) Industrial and Economic Impact • Creation of a scalable, exportable oncology platform with commercialization potential • Strengthening competitiveness of Korean SMEs in advanced biotechnology and AI for oncology Societal and Strategic Impact • Potential reduction in metastatic cancer mortality through earlier, function-driven intervention • Advancement of international, multi-institutional collaboration models and standards for precision oncology data and workflows

6. References

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